

Monday, Apr 22

Fixed Effects Approach

The fixed effects approach is to specify the many-leveled factor as we might normally do with a factor with fewer levels. The term “fixed effects” is used to distinguish it from the “random effects” approach which we will discuss later. The question then is if and how having such a factor compromises inferences.

Example: Consider again the `baserun` data.

```
library(dplyr)
library(tidyr)
baselong <- trtools::baserun %>% mutate(player = factor(letters[1:n()]))) %>%
  pivot_longer(cols = c(round, narrow, wide), names_to = "route", values_to = "time")
head(baselong)
```

```
# A tibble: 6 x 3
  player route    time
  <fct> <chr>   <dbl>
1 a     round    5.4
2 a     narrow   5.5
3 a     wide     5.55
4 b     round    5.85
5 b     narrow   5.7
6 b     wide     5.75
```

Consider a fixed effects model with an effect for player (but no interaction with route).

```
m.fix <- lm(time ~ route + player, data = baselong)
summary(m.fix)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	5.505e+00	0.05205	1.058e+02	1.320e-52
routeround	9.091e-03	0.02603	3.493e-01	7.286e-01
routewide	-7.500e-02	0.02603	-2.882e+00	6.208e-03
playerb	2.833e-01	0.07048	4.020e+00	2.366e-04
playerc	-5.000e-02	0.07048	-7.094e-01	4.820e-01
playerd	3.192e-15	0.07048	4.529e-14	1.000e+00
playere	3.333e-01	0.07048	4.729e+00	2.550e-05
playerf	5.000e-02	0.07048	7.094e-01	4.820e-01
playerg	-1.000e-01	0.07048	-1.419e+00	1.633e-01
playerh	-5.000e-02	0.07048	-7.094e-01	4.820e-01
playeri	-3.500e-01	0.07048	-4.966e+00	1.189e-05
playerj	3.000e-01	0.07048	4.256e+00	1.140e-04
playerk	-3.000e-01	0.07048	-4.256e+00	1.140e-04
playerl	6.667e-02	0.07048	9.459e-01	3.496e-01
playerm	-1.667e-02	0.07048	-2.365e-01	8.142e-01
playern	-4.833e-01	0.07048	-6.858e+00	2.323e-08
playero	-1.667e-02	0.07048	-2.365e-01	8.142e-01
playerp	1.667e-02	0.07048	2.365e-01	8.142e-01
playerq	2.866e-15	0.07048	4.067e-14	1.000e+00

```

playerr      1.667e-02    0.07048  2.365e-01  8.142e-01
players      -8.333e-02    0.07048 -1.182e+00  2.437e-01
playert       6.667e-02    0.07048  9.459e-01  3.496e-01
playeru       1.500e-01    0.07048  2.128e+00  3.923e-02
playerv       8.000e-01    0.07048  1.135e+01  2.238e-14

```

For comparison, we will also consider the marginal model using GEE, which should produce fairly accurate inferences.

```

library(geepack)
m.gee <- geeglm(time ~ route, data = baselong,
  id = player, corstr = "exchangeable")
trtools::lincon(m.gee) # easy way to get something like summary(m.gee)$coefficients

```

```

      estimate      se    lower    upper   tvalue df    pvalue
(Intercept)  5.534091 0.05411  5.42597  5.64221 102.2809 63 9.591e-72
routeround   0.009091 0.02564 -0.04215  0.06033   0.3546 63 7.241e-01
routewide   -0.075000 0.01839 -0.11176 -0.03824  -4.0775 63 1.301e-04

```

Here are the inferences for the expected time for each route, and the differences in the expected time between routes.

```

library(emmeans)
# Note: The player we choose does not matter.
pairs(emmeans(m.fix, ~route, at = list(player = "a")), infer = TRUE, adjust = "none")

```

```

contrast      estimate      SE df lower.CL upper.CL t.ratio p.value
narrow - round -0.00909 0.026 42  -0.0616   0.0434  -0.349  0.7286
narrow - wide   0.07500 0.026 42   0.0225   0.1275   2.882  0.0062
round - wide    0.08409 0.026 42   0.0316   0.1366   3.231  0.0024

```

Confidence level used: 0.95

```

pairs(emmeans(m.gee, ~route), infer = TRUE, adjust = "none")

```

```

contrast      estimate      SE df asymp.LCL asymp.UCL z.ratio p.value
narrow - round -0.00909 0.0256 Inf  -0.0593   0.0412  -0.355  0.7229
narrow - wide   0.07500 0.0184 Inf   0.0389   0.1110   4.077 <.0001
round - wide    0.08409 0.0307 Inf   0.0239   0.1443   2.737  0.0062

```

Confidence level used: 0.95

In *linear* models a fixed effects approach where the factor does not interact with other explanatory variables can produce valid inferences. But some inferences for explanatory variables that are confounded with the factor are not possible.

Example: Consider the following data on orthodontic measurements on children over time.

```

library(bayeslongitudinal)
head(Dental)

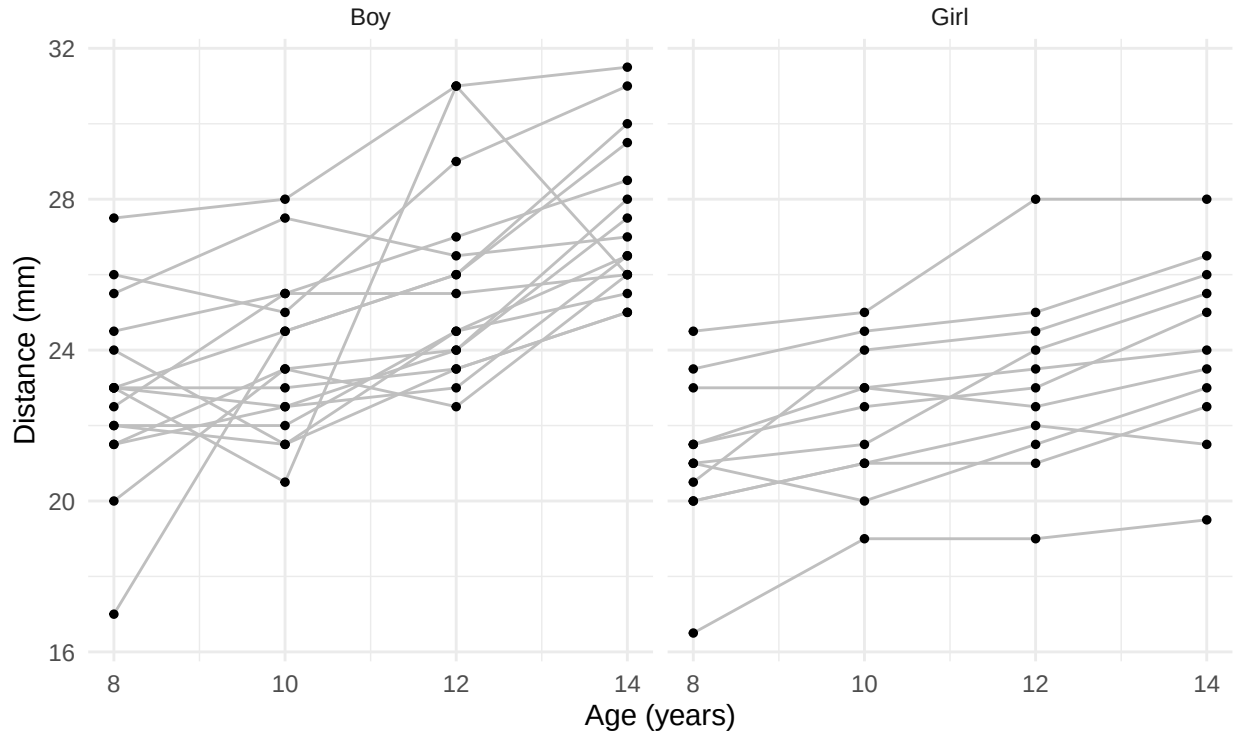
```

```

  gender id gencode distance age
1  Girl  1      1      21.0   8
2  Girl  1      1      20.0  10
3  Girl  1      1      21.5  12
4  Girl  1      1      23.0  14
5  Girl  2      1      21.0   8
6  Girl  2      1      21.5  10

```

```
p <- ggplot(Dental, aes(x = age, y = distance)) +
  geom_line(aes(group = id), color = grey(0.75)) +
  geom_point(size = 1) + facet_wrap(~ gender) +
  labs(x = "Age (years)", y = "Distance (mm)") + theme_minimal()
plot(p)
```



Age could be treated as a quantitative or categorical variable here. But the problem with the fixed effects approach is inferences for differences in expected distance between male and female children.

```
m.fix <- lm(distance ~ factor(id) + age + gender, data = Dental)
summary(m.fix)$coefficients
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	14.1130	0.98569	14.318	8.880e-24
factor(id)2	1.6250	1.01229	1.605	1.124e-01
factor(id)3	2.3750	1.01229	2.346	2.144e-02
factor(id)4	3.5000	1.01229	3.458	8.761e-04
factor(id)5	1.2500	1.01229	1.235	2.205e-01
factor(id)6	-0.2500	1.01229	-0.247	8.056e-01
factor(id)7	1.6250	1.01229	1.605	1.124e-01
factor(id)8	2.0000	1.01229	1.976	5.163e-02
factor(id)9	-0.2500	1.01229	-0.247	8.056e-01
factor(id)10	-2.8750	1.01229	-2.840	5.717e-03
factor(id)11	5.0000	1.01229	4.939	4.216e-06
factor(id)12	6.3750	1.01229	6.298	1.529e-08
factor(id)13	2.0000	1.01229	1.976	5.163e-02
factor(id)14	2.8750	1.01229	2.840	5.717e-03
factor(id)15	5.2500	1.01229	5.186	1.581e-06
factor(id)16	1.6250	1.01229	1.605	1.124e-01
factor(id)17	5.0000	1.01229	4.939	4.216e-06
factor(id)18	2.3750	1.01229	2.346	2.144e-02

```

factor(id)19  2.5000    1.01229    2.470 1.565e-02
factor(id)20  3.7500    1.01229    3.704 3.883e-04
factor(id)21  8.1250    1.01229    8.026 7.181e-12
factor(id)22  2.2500    1.01229    2.223 2.906e-02
factor(id)23  2.8750    1.01229    2.840 5.717e-03
factor(id)24  2.8750    1.01229    2.840 5.717e-03
factor(id)25  3.5000    1.01229    3.458 8.761e-04
factor(id)26  4.5000    1.01229    4.445 2.795e-05
factor(id)27  1.6250    1.01229    1.605 1.124e-01
age           0.6602    0.06161   10.716 3.952e-17

```

Notice that there is no indicator variable of gender! The `lm` function recognized that it is confounded with subject and removed it. We can see this if we construct a table of the number of observations by subject and sex.

```
with(Dental, table(id, gender))
```

```

      gender
id  Boy Girl
 1     0    4
 2     0    4
 3     0    4
 4     0    4
 5     0    4
 6     0    4
 7     0    4
 8     0    4
 9     0    4
10     0    4
11     0    4
12     4    0
13     4    0
14     4    0
15     4    0
16     4    0
17     4    0
18     4    0
19     4    0
20     4    0
21     4    0
22     4    0
23     4    0
24     4    0
25     4    0
26     4    0
27     4    0

```

These factors are *nested* (i.e., the variable `id` is nested in the variable `gender`).

By changing the order of the explanatory variables we can get sex in the model but then we lose a subject indicator variable.

```

m.fix <- lm(distance ~ age * gender + factor(id), data = Dental)
summary(m.fix)$coefficients

```

```

              Estimate Std. Error    t value Pr(>|t|)
(Intercept)  1.437e+01    1.0988   1.308e+01 1.834e-21

```

age	7.844e-01	0.0775	1.012e+01	6.442e-16
genderGirl	1.728e+00	1.6568	1.043e+00	3.001e-01
factor(id)2	1.625e+00	0.9803	1.658e+00	1.014e-01
factor(id)3	2.375e+00	0.9803	2.423e+00	1.770e-02
factor(id)4	3.500e+00	0.9803	3.570e+00	6.105e-04
factor(id)5	1.250e+00	0.9803	1.275e+00	2.060e-01
factor(id)6	-2.500e-01	0.9803	-2.550e-01	7.994e-01
factor(id)7	1.625e+00	0.9803	1.658e+00	1.014e-01
factor(id)8	2.000e+00	0.9803	2.040e+00	4.468e-02
factor(id)9	-2.500e-01	0.9803	-2.550e-01	7.994e-01
factor(id)10	-2.875e+00	0.9803	-2.933e+00	4.395e-03
factor(id)11	5.000e+00	0.9803	5.100e+00	2.274e-06
factor(id)12	4.750e+00	0.9803	4.845e+00	6.190e-06
factor(id)13	3.750e-01	0.9803	3.825e-01	7.031e-01
factor(id)14	1.250e+00	0.9803	1.275e+00	2.060e-01
factor(id)15	3.625e+00	0.9803	3.698e+00	3.999e-04
factor(id)16	7.863e-16	0.9803	8.021e-16	1.000e+00
factor(id)17	3.375e+00	0.9803	3.443e+00	9.237e-04
factor(id)18	7.500e-01	0.9803	7.651e-01	4.465e-01
factor(id)19	8.750e-01	0.9803	8.926e-01	3.748e-01
factor(id)20	2.125e+00	0.9803	2.168e+00	3.320e-02
factor(id)21	6.500e+00	0.9803	6.630e+00	3.768e-09
factor(id)22	6.250e-01	0.9803	6.375e-01	5.256e-01
factor(id)23	1.250e+00	0.9803	1.275e+00	2.060e-01
factor(id)24	1.250e+00	0.9803	1.275e+00	2.060e-01
factor(id)25	1.875e+00	0.9803	1.913e+00	5.942e-02
factor(id)26	2.875e+00	0.9803	2.933e+00	4.395e-03
age:genderGirl	-3.048e-01	0.1214	-2.511e+00	1.410e-02

If we wanted to compare the boys and girls, we could *in principle* estimate the average expected response for each sex, and the difference in these average expected responses (at a given age).

```
emmmeans(m.fix, ~ gender, at = list(age = 14))
```

gender	emmean	SE	df	lower.CL	upper.CL
Boy	27.3	0.29	79	26.7	27.9
Girl	24.1	0.35	79	23.4	24.8

Results are averaged over the levels of: id
Confidence level used: 0.95

```
pairs(emmeans(m.fix, ~ gender, at = list(age = 14)))
```

contrast	estimate	SE	df	t.ratio	p.value
Boy - Girl	3.24	0.454	79	7.122	<.0001

Results are averaged over the levels of: id

But there is maybe a limitation of such inferences — they are *for these particular children* (i.e., these 16 boys and 11 girls). We will see if/how we can generalize these inferences to other boys and girls of a given sex or age.

We also have a problem if we specify an interaction involving subject.

```
m.fix <- lm(distance ~ factor(id)*age + gender*age, data = Dental)
summary(m.fix)$coefficients
```

Estimate	Std. Error	t value	Pr(> t)
----------	------------	---------	----------

(Intercept)	1.725e+01	3.2882	5.246e+00	2.665e-06
factor(id)2	-3.050e+00	4.6502	-6.559e-01	5.147e-01
factor(id)3	-2.850e+00	4.6502	-6.129e-01	5.425e-01
factor(id)4	2.400e+00	4.6502	5.161e-01	6.079e-01
factor(id)5	2.350e+00	4.6502	5.054e-01	6.154e-01
factor(id)6	-2.500e-01	4.6502	-5.376e-02	9.573e-01
factor(id)7	-3.000e-01	4.6502	-6.451e-02	9.488e-01
factor(id)8	4.200e+00	4.6502	9.032e-01	3.704e-01
factor(id)9	8.500e-01	4.6502	1.828e-01	8.556e-01
factor(id)10	-3.700e+00	4.6502	-7.957e-01	4.297e-01
factor(id)11	1.700e+00	4.6502	3.656e-01	7.161e-01
factor(id)12	5.000e-02	4.6502	1.075e-02	9.915e-01
factor(id)13	-2.400e+00	4.6502	-5.161e-01	6.079e-01
factor(id)14	-1.250e+00	4.6502	-2.688e-01	7.891e-01
factor(id)15	7.450e+00	4.6502	1.602e+00	1.150e-01
factor(id)16	-3.600e+00	4.6502	-7.742e-01	4.422e-01
factor(id)17	1.700e+00	4.6502	3.656e-01	7.161e-01
factor(id)18	-2.300e+00	4.6502	-4.946e-01	6.229e-01
factor(id)19	2.500e+00	4.6502	5.376e-01	5.931e-01
factor(id)20	-2.850e+00	4.6502	-6.129e-01	5.425e-01
factor(id)21	4.000e+00	4.6502	8.602e-01	3.935e-01
factor(id)22	2.800e+00	4.6502	6.021e-01	5.496e-01
factor(id)23	-4.000e+00	4.6502	-8.602e-01	3.935e-01
factor(id)24	-1.445e+01	4.6502	-3.107e+00	3.008e-03
factor(id)25	1.850e+00	4.6502	3.978e-01	6.923e-01
factor(id)26	-3.750e+00	4.6502	-8.064e-01	4.235e-01
factor(id)27	-3.000e-01	4.6502	-6.451e-02	9.488e-01
age	3.750e-01	0.2929	1.280e+00	2.060e-01
factor(id)2:age	4.250e-01	0.4143	1.026e+00	3.095e-01
factor(id)3:age	4.750e-01	0.4143	1.147e+00	2.566e-01
factor(id)4:age	1.000e-01	0.4143	2.414e-01	8.102e-01
factor(id)5:age	-1.000e-01	0.4143	-2.414e-01	8.102e-01
factor(id)6:age	7.705e-15	0.4143	1.860e-14	1.000e+00
factor(id)7:age	1.750e-01	0.4143	4.224e-01	6.744e-01
factor(id)8:age	-2.000e-01	0.4143	-4.828e-01	6.312e-01
factor(id)9:age	-1.000e-01	0.4143	-2.414e-01	8.102e-01
factor(id)10:age	7.500e-02	0.4143	1.810e-01	8.570e-01
factor(id)11:age	3.000e-01	0.4143	7.242e-01	4.721e-01
factor(id)12:age	5.750e-01	0.4143	1.388e+00	1.708e-01
factor(id)13:age	4.000e-01	0.4143	9.656e-01	3.386e-01
factor(id)14:age	3.750e-01	0.4143	9.052e-01	3.694e-01
factor(id)15:age	-2.000e-01	0.4143	-4.828e-01	6.312e-01
factor(id)16:age	4.750e-01	0.4143	1.147e+00	2.566e-01
factor(id)17:age	3.000e-01	0.4143	7.242e-01	4.721e-01
factor(id)18:age	4.250e-01	0.4143	1.026e+00	3.095e-01
factor(id)19:age	6.119e-15	0.4143	1.477e-14	1.000e+00
factor(id)20:age	6.000e-01	0.4143	1.448e+00	1.533e-01
factor(id)21:age	3.750e-01	0.4143	9.052e-01	3.694e-01
factor(id)22:age	-5.000e-02	0.4143	-1.207e-01	9.044e-01
factor(id)23:age	6.250e-01	0.4143	1.509e+00	1.372e-01
factor(id)24:age	1.575e+00	0.4143	3.802e+00	3.672e-04
factor(id)25:age	1.500e-01	0.4143	3.621e-01	7.187e-01
factor(id)26:age	7.500e-01	0.4143	1.810e+00	7.580e-02
factor(id)27:age	1.750e-01	0.4143	4.224e-01	6.744e-01

Note that there are no terms for gender.

In general, the fixed effects approach does not permit proper inferences for explanatory variables that have a fixed effects variable nested within them. We are limited to inferences that are *crossed* with the fixed effect.

Fixed Effects and Nonlinear Models

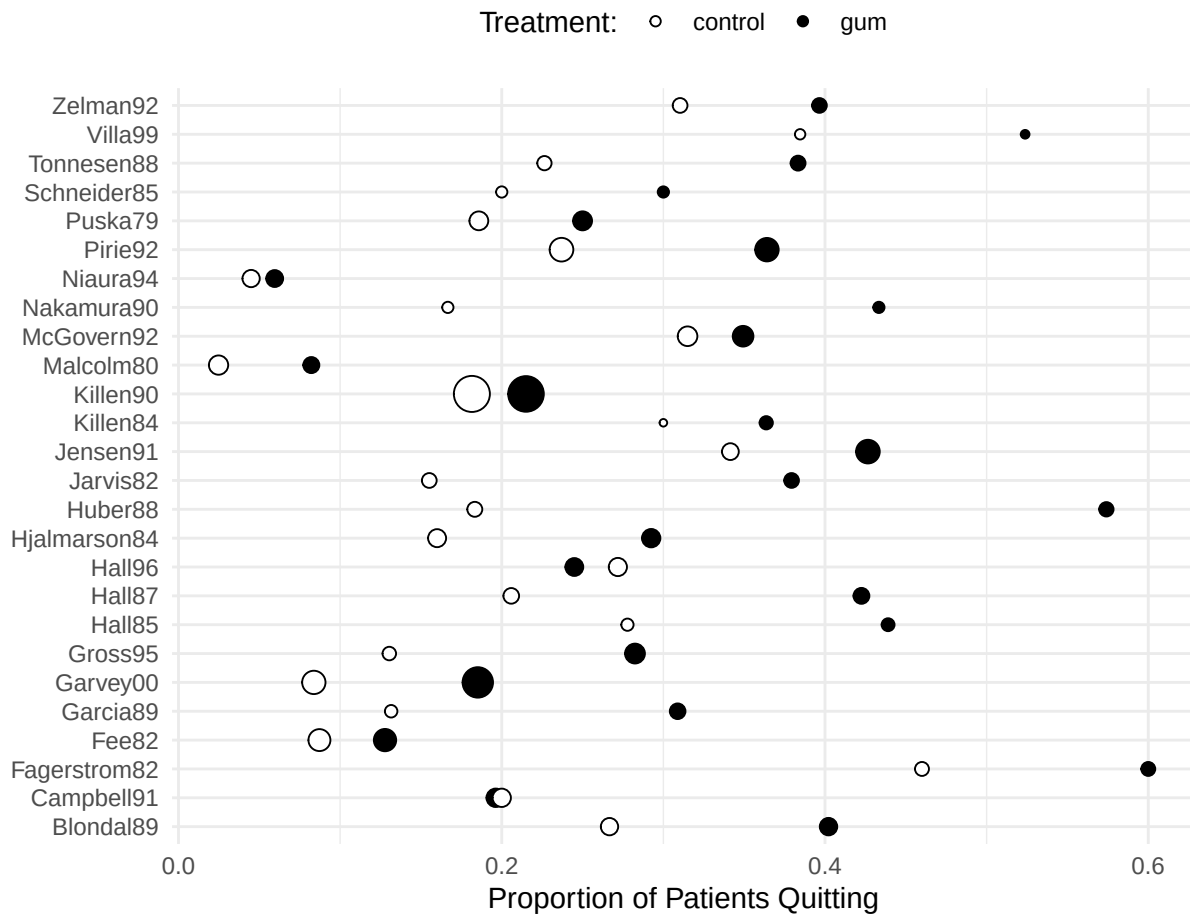
Fixed effects *can* produce valid inferences for nonlinear models (including generalized linear models), but not necessarily. It depends, in part, on the *number of parameters* relative to the number of observations.

Example: Recall the meta-analysis of 26 studies of the effect of nicotine gum on smoking cessation.

```
library(dplyr)
library(tidyr)
quitsmoke <- HSAUR3::smoking
quitsmoke$study <- rownames(quitsmoke)
quitsmoke.quits <- quitsmoke %>% dplyr::select(study, qt, qc) %>%
  rename(gum = qt, control = qc) %>%
  pivot_longer(cols = c(gum, control), names_to = "treatment", values_to = "quit")
quitsmoke.total <- quitsmoke %>% dplyr::select(study, tt, tc) %>%
  rename(gum = tt, control = tc) %>%
  pivot_longer(cols = c(gum, control), names_to = "treatment", values_to = "total")
quitsmoke <- full_join(quitsmoke.quits, quitsmoke.total) %>% mutate(study = factor(study)) %>% arrange(
head(quitsmoke)
```

```
# A tibble: 6 x 4
  study      treatment  quit total
  <fct>      <chr>      <int> <int>
1 Blondal89   gum           37    92
2 Blondal89   control       24    90
3 Campbell91  gum           21   107
4 Campbell91  control       21   105
5 Fagerstrom82 gum           30    50
6 Fagerstrom82 control      23    50
```

```
p <- ggplot(quitsmoke, aes(x = study, y = quit/total,
  size = total, fill = treatment)) +
  geom_point(pch = 21) + coord_flip() + guides(size = "none") +
  scale_fill_manual(values = c("White", "Black")) + theme_minimal() +
  labs(x = "", y = "Proportion of Patients Quitting", fill = "Treatment:") +
  theme(legend.position = "top")
plot(p)
```



Here is a fixed-effects logistic regression model.

```
m <- glm(cbind(quit, total-quit) ~ treatment + study,
  family = binomial, data = quitsmoke)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.95611	0.16223	-5.8935	3.782e-09
treatmentgum	0.51478	0.06571	7.8337	4.738e-15
studyCampbell91	-0.72182	0.23458	-3.0771	2.090e-03
studyFagerstrom82	0.82087	0.25660	3.1990	1.379e-03
studyFee82	-1.44471	0.23392	-6.1760	6.575e-10
studyGarcia89	-0.51371	0.27679	-1.8560	6.346e-02
studyGarvey00	-1.13119	0.19513	-5.7970	6.750e-09
studyGross95	-0.57476	0.23716	-2.4235	1.537e-02
studyHall85	0.11322	0.28635	0.3954	6.926e-01
studyHall87	-0.08874	0.24238	-0.3661	7.143e-01
studyHall96	-0.36356	0.22648	-1.6052	1.084e-01
studyHjalmarson84	-0.54554	0.23002	-2.3717	1.771e-02
studyHuber88	0.16466	0.25162	0.6544	5.128e-01
studyJarvis82	-0.32539	0.26384	-1.2333	2.175e-01
studyJensen91	0.18524	0.19887	0.9314	3.516e-01
studyKillen84	-0.05394	0.30863	-0.1748	8.613e-01
studyKillen90	-0.71634	0.17393	-4.1186	3.812e-05
studyMalcolm80	-2.28969	0.37670	-6.0784	1.214e-09
studyMcGovern92	-0.02349	0.20432	-0.1150	9.085e-01

studyNakamura90	-0.16186	0.32479	-0.4984	6.182e-01
studyNiaura94	-2.22602	0.37765	-5.8945	3.759e-09
studyPirie92	-0.15991	0.19132	-0.8358	4.033e-01
studyPuska79	-0.59867	0.22560	-2.6536	7.963e-03
studySchneider85	-0.41647	0.33913	-1.2281	2.194e-01
studyTonnesen88	-0.13127	0.25883	-0.5072	6.120e-01
studyVilla99	0.50932	0.33548	1.5182	1.290e-01
studyZelman92	0.08506	0.25163	0.3380	7.353e-01

We can estimate the odds ratio for the effect of treatment as follows.

```
rbind(pairs(emmeans(m, ~ treatment | study, type = "response"),
  reverse = TRUE), adjust = "none")
```

study	contrast	odds.ratio	SE	df	null	z.ratio	p.value
Blondal89	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Campbell191	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Fagerstrom82	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Fee82	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Garcia89	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Garvey00	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Gross95	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Hall85	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Hall87	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Hall96	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Hjalmarson84	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Huber88	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Jarvis82	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Jensen91	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Killen84	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Killen90	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Malcolm80	gum / control	1.67	0.11	Inf	1	7.834	<.0001
McGovern92	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Nakamura90	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Niaura94	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Pirie92	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Puska79	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Schneider85	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Tonnesen88	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Villa99	gum / control	1.67	0.11	Inf	1	7.834	<.0001
Zelman92	gum / control	1.67	0.11	Inf	1	7.834	<.0001

Tests are performed on the log odds ratio scale

Note that using `rbind` makes the output a bit more compact. Here is how we can do that using `contrast` from `trtools`.

```
trtools::contrast(m,
  a = list(treatment = "gum", study = unique(quitsmoke$study)),
  b = list(treatment = "control", study = unique(quitsmoke$study)),
  tf = exp, cnames = unique(quitsmoke$study))
```

	estimate	lower	upper
Blondal89	1.673	1.471	1.903
Campbell191	1.673	1.471	1.903
Fagerstrom82	1.673	1.471	1.903

Fee82	1.673	1.471	1.903
Garcia89	1.673	1.471	1.903
Garvey00	1.673	1.471	1.903
Gross95	1.673	1.471	1.903
Hall85	1.673	1.471	1.903
Hall87	1.673	1.471	1.903
Hall96	1.673	1.471	1.903
Hjalmarson84	1.673	1.471	1.903
Huber88	1.673	1.471	1.903
Jarvis82	1.673	1.471	1.903
Jensen91	1.673	1.471	1.903
Killen84	1.673	1.471	1.903
Killen90	1.673	1.471	1.903
Malcolm80	1.673	1.471	1.903
McGovern92	1.673	1.471	1.903
Nakamura90	1.673	1.471	1.903
Niaura94	1.673	1.471	1.903
Pirie92	1.673	1.471	1.903
Puska79	1.673	1.471	1.903
Schneider85	1.673	1.471	1.903
Tonnesen88	1.673	1.471	1.903
Villa99	1.673	1.471	1.903
Zelman92	1.673	1.471	1.903

Since the odds ratio is assumed to be the same for each study, we can just pick an arbitrary study.

```
pairs(emmeans(m, ~ treatment | study, type = "response",
  at = list(study = "Blondal89")), adjust = "none", reverse = TRUE)
```

```
study = Blondal89:
  contrast      odds.ratio    SE  df null z.ratio p.value
gum / control      1.67 0.11 Inf   1   7.834 <.0001
```

Tests are performed on the log odds ratio scale

```
trtools::contrast(m,
  a = list(treatment = "gum", study = "Blondal89"),
  b = list(treatment = "control", study = "Blondal89"),
  tf = exp)
```

```
estimate lower upper
1.673 1.471 1.903
```

Or we can omit study when using the **emmeans** package, causing it to average over study.

```
pairs(emmeans(m, ~ treatment, type = "response"), infer = TRUE, reverse = TRUE)
```

```
contrast      odds.ratio    SE  df asymp.LCL asymp.UCL null z.ratio p.value
gum / control      1.67 0.11 Inf   1.47      1.9    1   7.834 <.0001
```

Results are averaged over the levels of: study

Confidence level used: 0.95

Intervals are back-transformed from the log odds ratio scale

Tests are performed on the log odds ratio scale

Here is a model where the effect of nicotine gum varies over study.

```
m <- glm(cbind(quit, total-quit) ~ treatment * study,
  family = binomial, data = quitsmoke)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-1.011601	0.2384	-4.243904	2.197e-05
treatmentgum	0.615186	0.3194	1.925966	5.411e-02
studyCampbell91	-0.374693	0.3411	-1.098520	2.720e-01
studyFagerstrom82	0.851258	0.3706	2.297064	2.162e-02
studyFee82	-1.336595	0.3604	-3.709126	2.080e-04
studyGarcia89	-0.875469	0.5358	-1.633834	1.023e-01
studyGarvey00	-1.380932	0.3479	-3.969605	7.199e-05
studyGross95	-0.885519	0.4985	-1.776429	7.566e-02
studyHall85	0.056089	0.4419	0.126927	8.990e-01
studyHall87	-0.338326	0.3831	-0.883128	3.772e-01
studyHall96	0.026317	0.3254	0.080884	9.355e-01
studyHjalmarson84	-0.646627	0.3622	-1.785045	7.425e-02
studyHuber88	-0.482324	0.4100	-1.176276	2.395e-01
studyJarvis82	-0.682995	0.4340	-1.573798	1.155e-01
studyJensen91	0.354821	0.3332	1.064752	2.870e-01
studyKillen84	0.164303	0.5431	0.302551	7.622e-01
studyKillen90	-0.494459	0.2602	-1.899981	5.744e-02
studyMalcolm80	-2.660471	0.6314	-4.213818	2.511e-05
studyMcGovern92	0.234572	0.3055	0.767904	4.425e-01
studyNakamura90	-0.597837	0.5448	-1.097331	2.725e-01
studyNiaura94	-2.044756	0.5644	-3.622682	2.916e-04
studyPirie92	-0.157780	0.2881	-0.547567	5.840e-01
studyPuska79	-0.465665	0.3396	-1.371344	1.703e-01
studySchneider85	-0.374693	0.5149	-0.727661	4.668e-01
studyTonnesen88	-0.217065	0.4056	-0.535119	5.926e-01
studyVilla99	0.541597	0.4683	1.156483	2.475e-01
studyZelman92	0.213093	0.3706	0.574934	5.653e-01
treatmentgum:studyCampbell91	-0.638716	0.4699	-1.359285	1.741e-01
treatmentgum:studyFagerstrom82	-0.049378	0.5156	-0.095762	9.237e-01
treatmentgum:studyFee82	-0.187742	0.4742	-0.395873	6.922e-01
treatmentgum:studyGarcia89	0.466259	0.6334	0.736093	4.617e-01
treatmentgum:studyGarvey00	0.295743	0.4273	0.692111	4.889e-01
treatmentgum:studyGross95	0.349557	0.5756	0.607252	5.437e-01
treatmentgum:studyHall85	0.095203	0.5827	0.163387	8.702e-01
treatmentgum:studyHall87	0.422366	0.4997	0.845244	3.980e-01
treatmentgum:studyHall96	-0.755913	0.4542	-1.664447	9.602e-02
treatmentgum:studyHjalmarson84	0.159542	0.4712	0.338590	7.349e-01
treatmentgum:studyHuber88	1.177232	0.5377	2.189539	2.856e-02
treatmentgum:studyJarvis82	0.586934	0.5539	1.059684	2.893e-01
treatmentgum:studyJensen91	-0.254387	0.4191	-0.607000	5.439e-01
treatmentgum:studyKillen84	-0.327504	0.6621	-0.494666	6.208e-01
treatmentgum:studyKillen90	-0.404172	0.3504	-1.153314	2.488e-01
treatmentgum:studyMalcolm80	0.643954	0.7908	0.814266	4.155e-01
treatmentgum:studyMcGovern92	-0.460208	0.4107	-1.120609	2.625e-01
treatmentgum:studyNakamura90	0.725988	0.6912	1.050312	2.936e-01
treatmentgum:studyNiaura94	-0.318839	0.7592	-0.419943	6.745e-01
treatmentgum:studyPirie92	-0.003513	0.3863	-0.009096	9.927e-01
treatmentgum:studyPuska79	-0.236532	0.4544	-0.520520	6.027e-01
treatmentgum:studySchneider85	-0.076189	0.6849	-0.111241	9.114e-01

```
treatmentgum:studyTonnesen88    0.138056    0.5294  0.260782 7.943e-01
treatmentgum:studyVilla99      -0.049872    0.6749 -0.073900 9.411e-01
treatmentgum:studyZelman92     -0.236532    0.5046 -0.468741 6.393e-01
```

```
rbind(pairs(emmeans(m, ~ treatment | study, type = "response"),
  reverse = TRUE), adjust = "none")
```

study	contrast	odds.ratio	SE	df	null	z.ratio	p.value
Blondal89	gum / control	1.850	0.591	Inf	1	1.926	0.0541
Campbell191	gum / control	0.977	0.337	Inf	1	-0.068	0.9456
Fagerstrom82	gum / control	1.761	0.713	Inf	1	1.398	0.1622
Fee82	gum / control	1.533	0.537	Inf	1	1.219	0.2227
Garcia89	gum / control	2.949	1.613	Inf	1	1.977	0.0480
Garvey00	gum / control	2.487	0.706	Inf	1	3.209	0.0013
Gross95	gum / control	2.624	1.257	Inf	1	2.015	0.0440
Hall85	gum / control	2.035	0.992	Inf	1	1.458	0.1449
Hall87	gum / control	2.822	1.085	Inf	1	2.700	0.0069
Hall96	gum / control	0.869	0.281	Inf	1	-0.436	0.6629
Hjalmarson84	gum / control	2.170	0.752	Inf	1	2.236	0.0253
Huber88	gum / control	6.004	2.597	Inf	1	4.144	<.0001
Jarvis82	gum / control	3.327	1.506	Inf	1	2.657	0.0079
Jensen91	gum / control	1.434	0.389	Inf	1	1.330	0.1836
Killen84	gum / control	1.333	0.773	Inf	1	0.496	0.6198
Killen90	gum / control	1.235	0.178	Inf	1	1.464	0.1433
Malcolm80	gum / control	3.522	2.548	Inf	1	1.740	0.0818
McGovern92	gum / control	1.168	0.301	Inf	1	0.600	0.5482
Nakamura90	gum / control	3.824	2.344	Inf	1	2.188	0.0287
Niaura94	gum / control	1.345	0.926	Inf	1	0.430	0.6670
Pirie92	gum / control	1.844	0.400	Inf	1	2.816	0.0049
Puska79	gum / control	1.460	0.472	Inf	1	1.172	0.2414
Schneider85	gum / control	1.714	1.039	Inf	1	0.890	0.3737
Tonnesen88	gum / control	2.124	0.897	Inf	1	1.784	0.0744
Villa99	gum / control	1.760	1.046	Inf	1	0.951	0.3416
Zelman92	gum / control	1.460	0.571	Inf	1	0.969	0.3324

Tests are performed on the log odds ratio scale

The `contrast` function will let you estimate the *average* odds ratio (using the delta method).

```
trtools::contrast(m,
  a = list(treatment = "gum", study = unique(quitsmoke$study)),
  b = list(treatment = "control", study = unique(quitsmoke$study)),
  tf = function(x) mean(exp(x)))
```

estimate	se	lower	upper	tvalue	df	pvalue
2.14	0.228	1.693	2.587	9.383	Inf	6.431e-21

Note that the following will also “average” over study, but the averaging is done on the log odds scale, so the result is not actually the average odds ratio but instead the average logarithm of the odds ratio to which we have applied the exponential function, for which the interpretation is not clear.

```
pairs(emmeans(m, ~ treatment, type = "response"), infer = TRUE, reverse = TRUE)
```

contrast	odds.ratio	SE	df	asympt.LCL	asympt.UCL	null	z.ratio	p.value
gum / control	1.93	0.169	Inf	1.62	2.29	1	7.486	<.0001

Results are averaged over the levels of: study

Confidence level used: 0.95

Intervals are back-transformed from the log odds ratio scale

Tests are performed on the log odds ratio scale

But there is a way to estimate the average odds ratio using the **emmeans** package using **regrid**.

```
tmp <- pairs(emmeans(m, ~ treatment|study, type = "response"), reverse = TRUE)
emmeans(regrid(tmp), ~1)
```

	odds.ratio	SE	df	asympt.LCL	asympt.UCL
1					
overall	2.14	0.228	Inf	1.69	2.59

Results are averaged over the levels of: study

Confidence level used: 0.95

This does the same thing as using `tf = function(x) mean(exp(x))` with the **contrast** function.

These inferences are probably fine because while there can be a relatively large number of parameters, there are many observations per study as well. The data are *aggregated*. But we should note that when including the interaction the average odds ratio is *just for those studies*.

Where we can get into trouble is when there are only a few observations per level of the many-leveled factor.

The Incidental Parameter Problem and Fixed Effects Models

Example: Consider simulated data for a logistic regression model where we observe m observations of a binary response variable for each of n subjects. If we include a fixed effect for subject, the number of parameters is $1 + n$ and the number of binary observations is nm (m per subject). We will use a relatively large total sample size of $nm = 1000$, which *should* produce good estimates of the parameter for the effect of the explanatory variable, which has a value of $\beta_1 = 1$.

Here we have $n = 1000$ subjects with $m = 2$ observations per subject (1001 parameters).

```
set.seed(101)
n <- 1000
m <- 2
d <- data.frame(x = runif(n*m, -3, 3), z = rep(rnorm(n), each = m))
d$y <- rbinom(n*m, 1, plogis(d$x + d$z))
d$subject <- rep(1:n, each = m)

m <- glm(y ~ x + factor(subject), family = binomial, data = d)
```

Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred

```
head(summary(m)$coefficients)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	3.550	2.190e+00	1.620793	1.051e-01
x	2.026	1.264e-01	16.031488	7.702e-58
factor(subject)2	-26.387	1.246e+04	-0.002117	9.983e-01
factor(subject)3	-21.415	1.247e+04	-0.001718	9.986e-01
factor(subject)4	-24.302	1.135e+04	-0.002141	9.983e-01
factor(subject)5	-4.570	2.634e+00	-1.735034	8.273e-02

Here we have $n = 100$ subjects with $m = 20$ observations per subject (21 parameters).

```
set.seed(101)
n <- 100
m <- 20
d <- data.frame(x = runif(n*m, -3, 3), z = rep(rnorm(n), each = m))
```

```
d$y <- rbinom(n*m, 1, plogis(d$x + d$z))
d$subject <- rep(1:n, each = m)

m <- glm(y ~ x + factor(subject), family = binomial, data = d)
head(summary(m)$coefficients)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	0.5058	0.56849	0.8898	3.736e-01
x	1.0706	0.04924	21.7399	8.601e-105
factor(subject)2	-4.0151	1.00050	-4.0131	5.994e-05
factor(subject)3	-0.9076	0.78983	-1.1492	2.505e-01
factor(subject)4	0.5687	0.92331	0.6159	5.379e-01
factor(subject)5	-1.5492	0.86751	-1.7858	7.413e-02

Having too many parameters relative to the number of observations causes problems.

Conditional Maximum Likelihood

In some models (namely logistic and Poisson regression), we can handle the incidental parameter problem if it only involves only a “main effect” by using what is called a *conditional likelihood* which in a sense removes the effect of the factor. Consider again our data with $n = 1000$ subjects and $m = 2$ binary observations per subject.

```
set.seed(101)
n <- 1000
m <- 2
d <- data.frame(x = runif(n*m, -3, 3), z = rep(rnorm(n), each = m))
d$y <- rbinom(n*m, 1, plogis(d$x + d$z))
d$subject <- rep(1:n, each = m)

library(survival) # for the clogit function
m <- clogit(y ~ x + strata(subject), data = d)
summary(m)
```

Call:

```
coxph(formula = Surv(rep(1, 2000L), y) ~ x + strata(subject),
      data = d, method = "exact")
```

n= 2000, number of events= 982

	coef	exp(coef)	se(coef)	z	Pr(> z)
x	1.0132	2.7544	0.0894	11.3	<2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

	exp(coef)	exp(-coef)	lower .95	upper .95
x	2.75	0.363	2.31	3.28

Concordance= 0.861 (se = 0.023)

Likelihood ratio test= 335 on 1 df, p=<2e-16

Wald test = 128 on 1 df, p=<2e-16

Score (logrank) test = 255 on 1 df, p=<2e-16

The *clogit* function requires that the response is *binary*, so to apply it to the smoking cessation data we would need to reformat the data.

```
quitsmoke <- quitsmoke %>%
  mutate(noquit = total - quit) %>% dplyr::select(-total) %>%
  pivot_longer(cols = c(quit, noquit), names_to = "outcome", values_to = "count")
head(quitsmoke)
```

```
# A tibble: 6 x 4
  study      treatment outcome count
  <fct>      <chr>      <chr>  <int>
1 Blondal89 gum        quit     37
2 Blondal89 gum        noquit   55
3 Blondal89 control    quit     24
4 Blondal89 control    noquit   66
5 Campbell191 gum      quit     21
6 Campbell191 gum      noquit   86
```

```
quitsmoke <- quitsmoke %>% uncount(count) %>%
  mutate(y = ifelse(outcome == "quit", 1, 0))
head(quitsmoke)
```

```
# A tibble: 6 x 4
  study      treatment outcome      y
  <fct>      <chr>      <chr>  <dbl>
1 Blondal89 gum        quit      1
2 Blondal89 gum        quit      1
3 Blondal89 gum        quit      1
4 Blondal89 gum        quit      1
5 Blondal89 gum        quit      1
6 Blondal89 gum        quit      1
```

```
m <- clogit(y ~ treatment + strata(study), data = quitsmoke)
summary(m)
```

Call:

```
coxph(formula = Surv(rep(1, 5846L), y) ~ treatment + strata(study),
      data = quitsmoke, method = "exact")
```

n= 5846, number of events= 1394

```
              coef exp(coef) se(coef)      z Pr(>|z|)
treatmentgum 0.5123   1.6691   0.0656  7.81 5.5e-15 ***
---
```

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```
              exp(coef) exp(-coef) lower .95 upper .95
treatmentgum      1.67      0.599    1.47      1.9
```

```
Concordance= 0.545 (se = 0.011 )
Likelihood ratio test= 62.3 on 1 df,  p=3e-15
Wald test               = 61.1 on 1 df,  p=6e-15
Score (logrank) test = 61.7 on 1 df,  p=4e-15
```

Poisson regression is an interesting special case when using either a fixed effects approach or conditional maximum likelihood. Here the two approaches produce the same results.

Example: Consider the following data from a case-control study that compared the number of *naevi* between children with (case) and without (control) spina bifida.

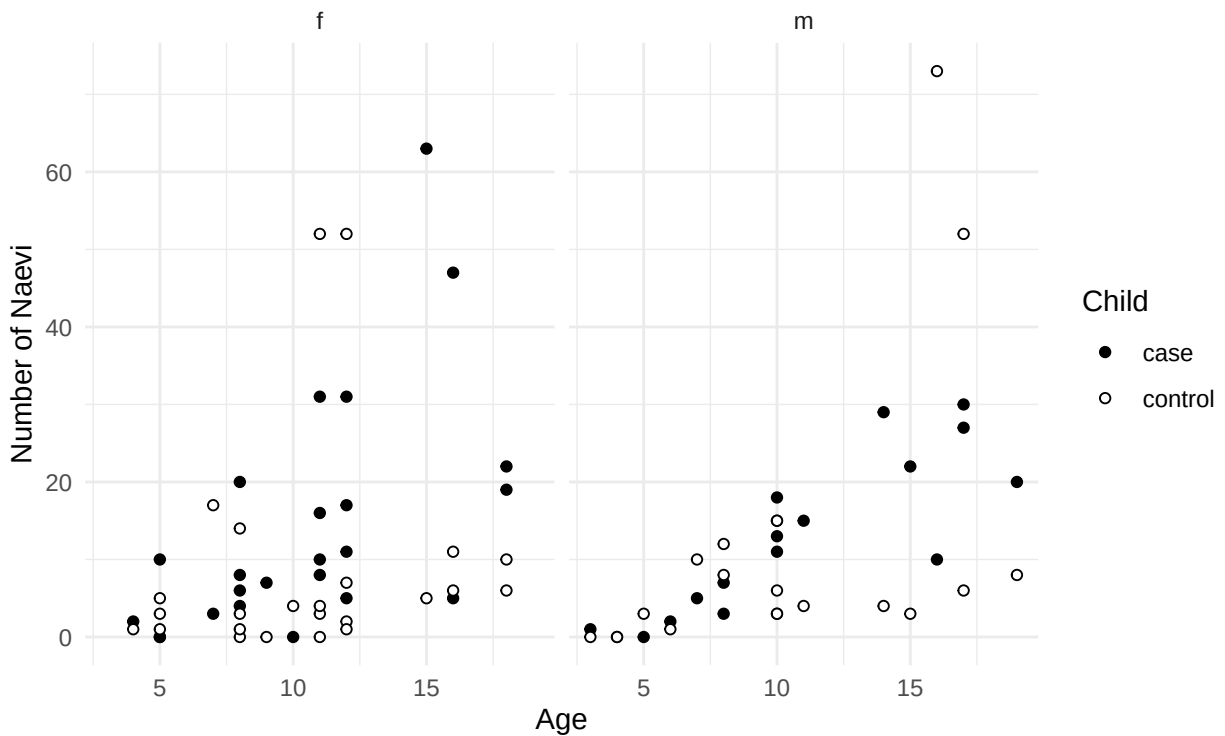
```
library(dplyr)
library(tidyr)
library(trtools) # for the naevi data
naevi$set <- factor(1:nrow(naevi)) # data frame naevi is from trtools package
head(naevi)
```

```
  sex age case control set
1  f  16    5        6    1
2  f   5    0        3    2
3  m  10   15       15    3
4  m   6    2        1    4
5  f  12   11        7    5
6  f  18   22        6    6
```

```
naevilong <- naevi %>% pivot_longer(cols = c(case, control),
  names_to = "child", values_to = "count")
head(naevilong)
```

```
# A tibble: 6 x 5
  sex    age set  child  count
<fct> <int> <fct> <chr>  <int>
1  f      16  1    case     5
2  f      16  1   control    6
3  f       5  2    case     0
4  f       5  2   control    3
5  m      10  3    case    15
6  m      10  3   control    15
```

```
p <- ggplot(naevilong, aes(x = age, y = count, fill = child)) +
  facet_wrap(~ sex) + geom_point(shape = 21) +
  scale_fill_manual(values = c("black", "white")) +
  labs(x = "Age", y = "Number of Naevi", fill = "Child") + theme_minimal()
plot(p)
```

The children have been matched by age and sex. But there may be other variables that are correlated with age and sex that are also related to the number of naevi, and these will potential cause a “set effect” on the counts. There are several ways we could handle this.

```
m <- glm(count ~ child + set, family = poisson, data = naevilong)
head(summary(m)$coefficients)
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.8491	0.30273	6.108	1.009e-09
childcontrol	-0.3130	0.06428	-4.870	1.118e-06
set2	-1.2993	0.65134	-1.995	4.607e-02
set3	1.0033	0.35248	2.846	4.422e-03
set4	-1.2993	0.65134	-1.995	4.607e-02
set5	0.4925	0.38271	1.287	1.982e-01

Note that we omit age and sex since those variables vary between but not within sets and are thus “redundant” with the effect of set (if you include them it will not change inferences concerning the effect of child). Let’s estimate the effect of being a case.

```
trtools::contrast(m, tf = exp,
  a = list(child = "case", set = "1"),
  b = list(child = "control", set = "1"))
```

```
estimate lower upper
1.368 1.206 1.551
```

Note that the set does not matter.

There is a trick to using conditional maximum likelihood here. It can be done by using logistic regression.

```
m <- glm(cbind(case, control) ~ 1, family = binomial, data = naevi)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	1.368	0.192	7.125	1.000e-12

```
(Intercept)    0.313    0.06428    4.87 1.118e-06
```

Strange model. But look at this.

```
trtools::lincon(m, tf = exp)
```

```
              estimate lower upper
(Intercept)    1.368 1.206 1.551
```

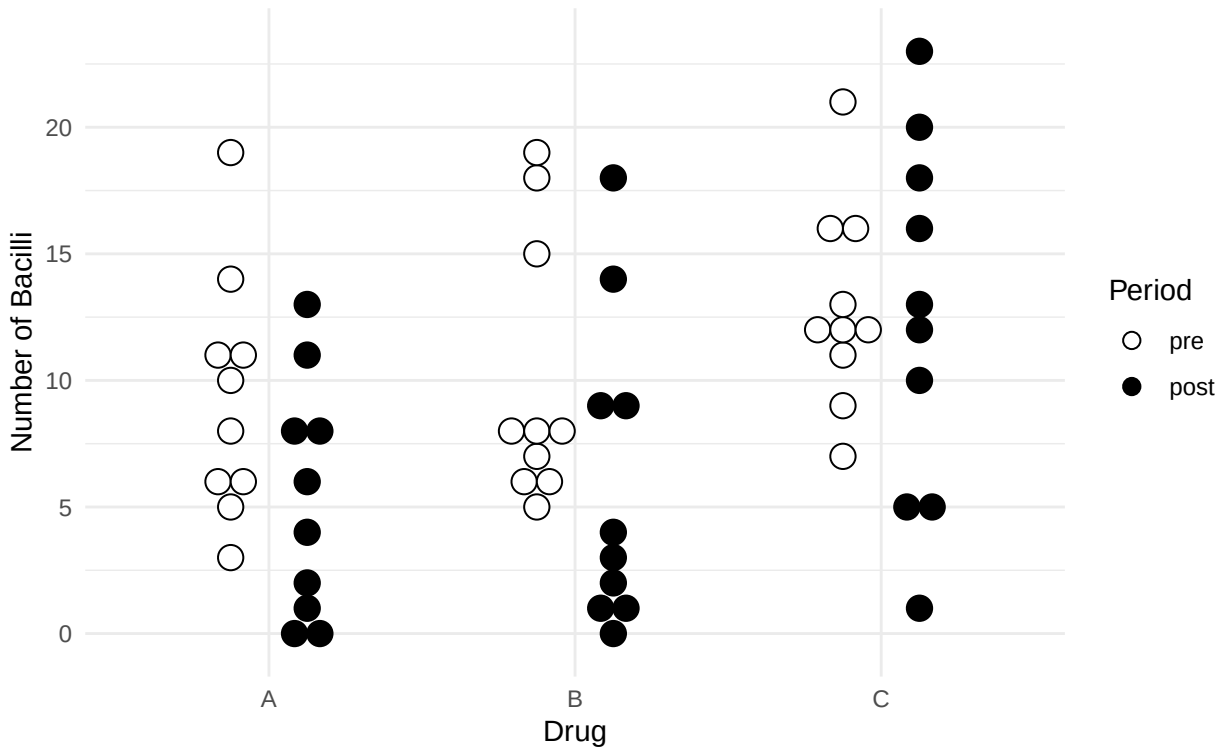
There's maybe no real advantage to using conditional maximum likelihood here via logistic regression except that in problems with *many* levels it is computationally faster.

Example: Consider data from a study of the effect of three antibiotics on leprosy bacilli. Note that if you want to install **ALA** you will need to use `install.packages("ALA", repos = "http://R-Forge.R-project.org")` because it is not kept on the default repository.

```
library(ALA)
head(leprosy)
```

```
   id drug period nBacilli
1   1   A    pre        11
31  1   A   post         6
2   2   B    pre         6
32  2   B   post         0
3   3   C    pre        16
33  3   C   post        13
```

```
p <- ggplot(leprosy, aes(x = drug, y = nBacilli, fill = period)) +
  geom_dotplot(binaxis = "y", method = "histodot",
    stackdir = "center", binwidth = 1,
    position = position_dodge(width = 0.5)) +
  scale_fill_manual(values = c("white", "black")) +
  labs(x = "Drug", y = "Number of Bacilli", fill = "Period") +
  theme_minimal()
plot(p)
```



First a fixed effects approach.

```
m <- glm(nBacilli ~ factor(id) + drug*period, family = poisson, data = leprosy)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	2.38221	0.2505	9.51158	1.878e-21
factor(id)2	-1.06668	0.4829	-2.20896	2.718e-02
factor(id)3	0.31547	0.3178	0.99269	3.209e-01
factor(id)4	-0.75377	0.4287	-1.75809	7.873e-02
factor(id)5	-0.77900	0.4376	-1.78006	7.507e-02
factor(id)6	0.08367	0.3316	0.25229	8.008e-01
factor(id)7	-0.88730	0.4491	-1.97579	4.818e-02
factor(id)8	-0.55586	0.4081	-1.36218	1.731e-01
factor(id)9	0.31547	0.3178	0.99269	3.209e-01
factor(id)10	0.25783	0.3229	0.79843	4.246e-01
factor(id)11	-0.66122	0.4215	-1.56888	1.167e-01
factor(id)12	-0.41277	0.3714	-1.11137	2.664e-01
factor(id)13	0.56798	0.3036	1.87099	6.135e-02
factor(id)14	0.72508	0.3071	2.36126	1.821e-02
factor(id)15	0.73237	0.2987	2.45163	1.422e-02
factor(id)16	-0.53063	0.3985	-1.33147	1.830e-01
factor(id)17	-0.37353	0.3871	-0.96495	3.346e-01
factor(id)18	0.28038	0.3197	0.87694	3.805e-01
factor(id)19	0.30228	0.3198	0.94508	3.446e-01
factor(id)20	0.63807	0.3112	2.05063	4.030e-02
factor(id)21	-0.21861	0.3540	-0.61750	5.369e-01
factor(id)22	-0.88730	0.4491	-1.97579	4.818e-02
factor(id)23	-0.02523	0.3540	-0.07126	9.432e-01
factor(id)24	0.28038	0.3197	0.87694	3.805e-01
factor(id)25	0.11123	0.3338	0.33316	7.390e-01

```

factor(id)26      -1.06668      0.4829 -2.20895 2.718e-02
factor(id)27      -0.97238      0.4376 -2.22198 2.628e-02
factor(id)28      -1.73460      0.6262 -2.76994 5.607e-03
factor(id)29       0.31961      0.3289  0.97173 3.312e-01
factor(id)30       0.41391      0.3127  1.32381 1.856e-01
periodpost       -0.56231      0.1721 -3.26721 1.086e-03
drugB:periodpost  0.06801      0.2367  0.28736 7.738e-01
drugC:periodpost  0.51468      0.2133  2.41279 1.583e-02

```

Now we can estimate the rate ratio for the effect of period for each drug.

```

pairs(emmeans(m, ~ period | drug, type = "response"),
      reverse = TRUE, infer = TRUE)

```

```

drug = A:
  contrast      ratio      SE df asymp.LCL asymp.UCL null z.ratio p.value
post / pre 0.570 0.0981 Inf      0.407      0.798  1 -3.267 0.0011

```

```

drug = B:
  contrast      ratio      SE df asymp.LCL asymp.UCL null z.ratio p.value
post / pre 0.610 0.0991 Inf      0.444      0.839  1 -3.043 0.0023

```

```

drug = C:
  contrast      ratio      SE df asymp.LCL asymp.UCL null z.ratio p.value
post / pre 0.954 0.1202 Inf      0.745      1.221  1 -0.378 0.7055

```

Results are averaged over the levels of: id

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale

Interestingly for this particular model we could actually drop `factor(id)` from the model entirely as it is nested with drug. We would obtain the same inferences! But do not assume that this is the case in general.

Note how `rbind` makes the output a bit more compact. Nice feature.

```

rbind(pairs(emmeans(m, ~ period | drug, type = "response"),
              reverse = TRUE, infer = TRUE), adjust = "none")

```

```

drug contrast      ratio      SE df asymp.LCL asymp.UCL null z.ratio p.value
A  post / pre 0.570 0.0981 Inf      0.407      0.798  1 -3.267 0.0011
B  post / pre 0.610 0.0991 Inf      0.444      0.839  1 -3.043 0.0023
C  post / pre 0.954 0.1202 Inf      0.745      1.221  1 -0.378 0.7055

```

Results are averaged over some or all of the levels of: id

Confidence level used: 0.95

Intervals are back-transformed from the log scale

Tests are performed on the log scale

How do we compare the rate ratios between drugs? Here are a couple of approaches.

```

pairs(pairs(emmeans(m, ~ period | drug, type = "response"),
              reverse = TRUE), by = NULL, adjust = "none")

```

```

contrast      ratio      SE df null z.ratio p.value
(post / pre A) / (post / pre B) 0.934 0.221 Inf  1 -0.287 0.7738
(post / pre A) / (post / pre C) 0.598 0.128 Inf  1 -2.413 0.0158
(post / pre B) / (post / pre C) 0.640 0.132 Inf  1 -2.172 0.0298

```

Results are averaged over the levels of: id
 Tests are performed on the log scale

```
pairs(rbind(pairs(emmeans(m, ~ period | drug, type = "response"),
  reverse = TRUE)), adjust = "none")
```

contrast	ratio	SE	df	null	z.ratio	p.value
(A post / pre) / (B post / pre)	0.934	0.221	Inf	1	-0.287	0.7738
(A post / pre) / (C post / pre)	0.598	0.128	Inf	1	-2.413	0.0158
(B post / pre) / (C post / pre)	0.640	0.132	Inf	1	-2.172	0.0298

Results are averaged over some or all of the levels of: id
 Tests are performed on the log scale

Now consider conditional maximum likelihood using logistic regression.

```
leprosylong <- leprosy %>%
  pivot_wider(names_from = "period", values_from = "nBacilli")
head(leprosylong)
```

```
# A tibble: 6 x 4
  id   drug   pre post
<fct> <fct> <int> <int>
1 1     A     11     6
2 2     B      6     0
3 3     C     16    13
4 4     A      8     0
5 5     B      6     2
6 6     C     13    10
```

```
m <- glm(cbind(post, pre) ~ drug, family = binomial, data = leprosylong)
summary(m)$coefficients
```

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-0.56231	0.1721	-3.2672	0.001086
drugB	0.06801	0.2367	0.2874	0.773834
drugC	0.51468	0.2133	2.4128	0.015831

Our estimates of the “odds” of a bacilli in the post period equals the estimated rate ratio for the effect of a drug.

```
trtools::contrast(m, tf = exp,
  a = list(drug = c("A", "B", "C")), cnames = c("A", "B", "C"))
```

	estimate	lower	upper
A	0.5699	0.4067	0.7985
B	0.6100	0.4437	0.8387
C	0.9535	0.7448	1.2206

When there are more than two observations per level, conditional maximum likelihood can be done using a multinomial logistic regression model. But there’s no advantage to using conditional maximum likelihood here either since we can get the same results using a more straightforward fixed effects approach.

Limitations of the Fixed Effects Approach

1. Some inferences may be impossible. Meaningful inferences are largely limited to explanatory variables that are *crossed* with the fixed effect.
2. Possibly poor inferences in nonlinear or generalized linear models.

3. More computationally intensive (although there are solutions).